Practice workbook · Angle between a line and a plane

Finding the angle between a line and a plane by dropping a perpendicular to the plane and using the line's projection.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-15

1. Question 4 · part whole · 6 marks

2024 NOV · Paper 01

Worked-example retry · Attempt part whole only.

4

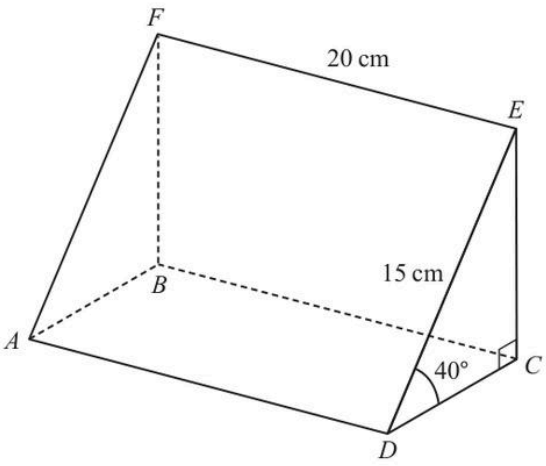


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right prism $ABCDEF$
Triangle CDE is a cross section of the prism.

- $\angle DCE = \angle ABF = 90°$
- $\angle EDC = \angle FAB = 40°$
- $DE = AF = 15\,\text{cm}$
- $EF = DA = CB = 20\,\text{cm}$

Find, in degrees to one decimal place, the size of the angle between the line FD and the plane $DCBA$

(6)

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1. Question 4 · part whole · 6 marks

2024 NOV · Paper 01

Worked-example retry · Attempt part whole only.

Question 4 continued

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(Total for Question 4 is 6 marks)

1. Question 4 · part whole · 6 marks · Working and answer

Working and answer

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2. Question 3 · part b · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

3

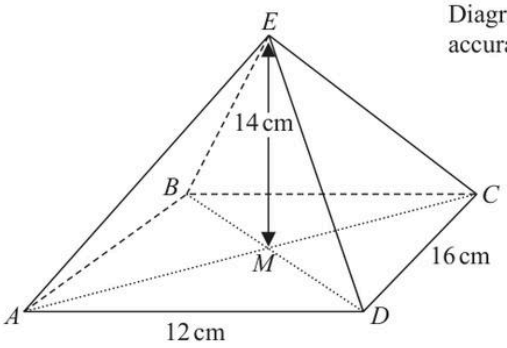


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the right pyramid $ABCDE$. The base, $ABCD$, of the pyramid is a horizontal rectangle with $AD = 12$ cm and $CD = 16$ cm. The height ME of the pyramid is 14 cm where M is the point of intersection of the diagonals of the base.

The sloping edges, EA, EB, EC and ED of the pyramid are all of equal length.

(a) Calculate, to 3 significant figures, the length of a sloping edge. **(3)**

Calculate, in degrees to one decimal place, the size of

(b) the angle between AE and the base, **(3)**

(c) the angle between the plane AED and the base. **(3)**

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P 6 2 2 8 0 A 0 6 3 2

2. Question 3 · part b · 3 marks · Working and answer

Working and answer

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3. Question 5 · part b · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

5

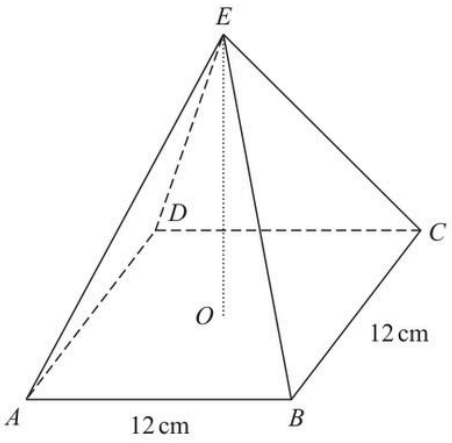


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a right pyramid with a square base $ABCD$ and vertex E.

The base of the pyramid is horizontal with $AB = BC = 12$ cm.
The diagonals of the base intersect at the point O.

The vertex E of the pyramid is vertically above O and the angle between EA and the plane $ABCD$ is 30°

The height of the pyramid is h cm.

(a) Find the exact value of h

(3)

The point F lies on AD such that $AF : FD = 1 : 4$

(b) Calculate, to the nearest degree, the size of the angle EFO.

(4)

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3. Question 5 · part b · 4 marks

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Further practice · Attempt part b only.

Question 5 continued

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3. Question 5 · part b · 4 marks

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Further practice · Attempt part b only.

Question 5 continued

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3. Question 5 · part b · 4 marks

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Further practice · Attempt part b only.



Question 5 continued

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(Total for Question 5 is 7 marks)

3. Question 5 · part b · 4 marks · Working and answer

Working and answer

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4. Question 9 · part d · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part d only.

9

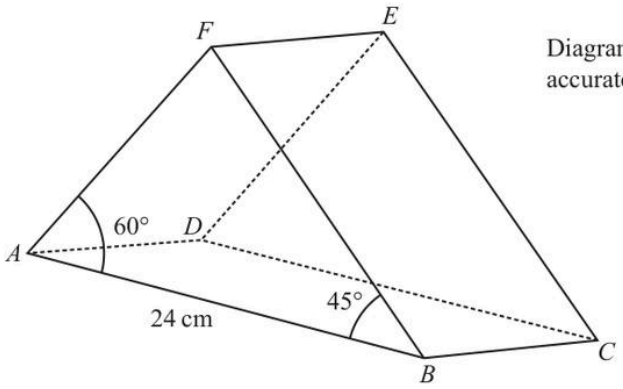


Figure 5

Figure 5 shows a right triangular prism $ABCDEF$ where $ABCD$ is a rectangle.

$AF = DE$ $BF = CE$ $AD = FE = BC$ $AB = DC = 24$ cm

$\angle ABF = \angle DCE = 45°$ $\angle BAF = \angle CDE = 60°$

Using a formula from page 2,

(a) show that $\sin AFB = \dfrac{\sqrt{2} + \sqrt{6}}{4}$

(3)

Without using a calculator,

(b) show that $BF = 12\left(3\sqrt{2} - \sqrt{6}\right)$ cm

(5)

The angle between the plane AEB and the plane $ABCD$ is 65°

(c) Find, in cm to 2 significant figures, the length of EF

(3)

(d) Find, in degrees to one decimal place, the size of the angle between the line CF and the plane $ABCD$

(4)

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4. Question 9 · part d · 4 marks

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Further practice · Attempt part d only.

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Question 9 continued

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4. Question 9 · part d · 4 marks

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Further practice · Attempt part d only.

Question 9 continued

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4. Question 9 · part d · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part d only.

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Question 9 continued

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(Total for Question 9 is 15 marks)

4. Question 9 · part d · 4 marks · Working and answer

Working and answer

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5. Question 11 · part b · 2 marks

2025 NOV · Paper 02

Further practice · Attempt part b only.

11

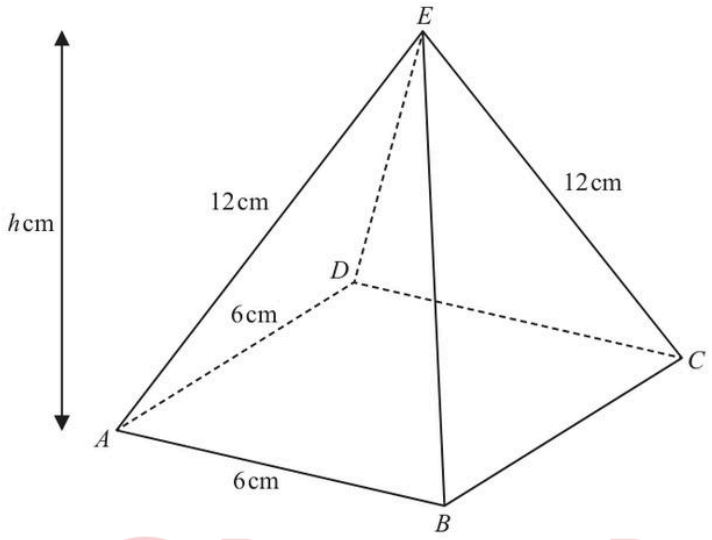


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a right pyramid $ABCDE$ with

- vertex E
- horizontal square base $ABCD$ of side 6 cm
- $EA = EB = EC = ED = 12$ cm
- vertical height of h cm

(a) Show that $h = 3\sqrt{14}$

(3)

(b) Find, in degrees to one decimal place, the size of the angle between EB and the plane $ABCD$

(2)

The midpoint of AE is P and the midpoint of BE is Q

(c) Find, to the nearest degree, the size of the obtuse angle between the plane EPQ and the plane $PQCD$

(6)

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5. Question 11 · part b · 2 marks · Working and answer

Working and answer

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